SAPTHAGIRI NPS UNIVERSITY
UNMATCHED EXCELLENCE,UNLIMITED POTENTIAL

School of Engineering and Technology

SRN: | | | | | | | | | | | | | |

Internal Assessment – 1 (CIA-1)

Department: Physics | **Date:** 09/12/2025

Course Name: Engineering Physics | **Course Code:** 25BEPHY102

Batch: 2025-26 | **Max. Marks:** 50

Semester: I | **Duration:** 90 Mins

Note: Answer all the questions. Q1 & Q2 carries 10 Marks each, Q3 - Q6 carry 20 Marks each

Q No	Question	Marks	RBT Level	CO
1.	a. What is a matter wave? Explain the de Broglie wavelength.	5	1	1
	b. What are qubits? List five properties of qubits.	5	1	2
	OR			
2.	a. State and explain Heisenberg's uncertainty principle.	5	2	1
	b. What is a unitary matrix? Give an example.	5	2	2
3.	a. Derive an expression for the de Broglie wavelength in terms of kinetic energy. Determine the wavelength of an electron accelerated by a potential V	10	3	1
	b. Develop Schrödinger's time-independent equation for one dimension.	10	2	
	OR			
4.	a. Using the Schrödinger wave equation, derive the expression for the energy eigenvalues for the first three energy states and the Eigen functions of a particle in a one-dimensional box.	10	3	1
	b. An electron has a speed of 4.8 x 10^5 m/s accurate to 0.012%. With what accuracy can the position of electron be located?	5	3	
	c. An electron is bound in 1-D potential well of width 0.18nm, find its energy value in eV in the second excited state.	5	3	
5.	a. State and explain Moore's law with graph and discuss its failure.	5	2	2
	b. Give the Definition, matrix form, operations and truth table for. Pauli's X gate, Y gate and Z gate.	10	2	
	c. Verify the given matrix is unitary or not $A = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$	5	3	
	OR			
6.	a. Explain the representation of Qubits. Give the matrix representation for single qubit and 2 qubits.	5	1	2
	b. Give the symbol, truth table, and matrix for the controlled-NOT (CNOT) gate. Explain the operation for each possible input.	10	2	
	c. Define the transpose and the conjugate of a matrix, with an example for each.	5	1	

Course outcomes: Students will be able to

CO1: Use the quantum mechanical laws of physics and solve basic problems.

CO2: Explore the applications of quantum mechanics in quantum computing.

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